

ABHINEET MENON

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PROFESSIONAL SUMMARY

AI & Software Engineer specializing in LLM orchestration, RAG, and multi-agent frameworks. Combines academic research in model evaluation with enterprise experience engineering scalable data pipelines.

EDUCATION

Master of Science in Computer Science (GPA 4.0/4.0)

Aug 2024 - May 2026

University of Illinois Chicago

Chicago, IL

PROFESSIONAL EXPERIENCE

Engineer | *LTIMindtree, Mumbai*

Jun 2022 - Mar 2024

- Engineered scalable ETL pipelines to migrate high-volume data to Snowflake, establishing the optimized data architecture required for high-throughput downstream analytics.
- Optimized feature extraction and data retrieval processes by refining Snowflake query logic, reducing upstream data latency by up to 40%.
- Led a 10-member intern team and drove agile scrum meetings to ensure alignment with stakeholder requirements.
- Developed incremental-load pipelines in Azure Data Factory and integrated Power BI dashboards, enabling real-time data insights for business users.

Web Development Intern | *Abner Security, Kokata*

May 2021 - Sep 2021

- Reengineered the company's WordPress-based web platform, implementing responsive layouts, performance optimization, and SEO best practices, increasing organic traffic by 30% and reducing average page load time.
- Customized and integrated third-party plugins to enhance site functionality, security, and analytics, ensuring maintainability and extensibility of the codebase.

Web Developer Intern | *Innovatiive Creators, Mumbai*

Oct 2020 - Nov 2020

- Developed a dynamic PHP-based admin panel for the National Institute of Hospitality Management, streamlining content management workflows and reducing update time by 40%.
- Designed and implemented modules for real-time traffic analytics, student enrollment tracking, and lead management, consolidating core operational data into a unified backend interface.

PROJECTS

CaseQuery - Legal RAG System | *Python, FAISS, Llama-3, GitHub, REST APIs*

Nov 2025 - Dec 2025

- Architected a Retrieval-Augmented Generation (RAG) system for legal QA, integrating a hybrid retrieval pipeline to reduce LLM or AI hallucinations by anchoring answers to specific case law citations.
- Implemented embeddings generation and document chunking strategies for dense and sparse retrieval with a custom evaluation pipeline, achieving a 50% improvement in ROUGE-L scores over zero-shot baselines, while implementing rigorous evaluation to guarantee citation accuracy.
- Built an interactive frontend and modular backend to process thousands of legal documents, utilizing concurrent API calls to OpenRouter for low-latency inference.

MathSCV - Multi-Agent LLM Debate System | *Python, DeepSeek API, Prompt Engineering*

Apr 2026 - May 2026

- Engineered a multi-agent LLM pipeline (Solver, Critic, Verifier) utilizing DeepSeek-V4-Flash to evaluate mathematical reasoning capabilities across GSM8K, SVAMP, and ASDiv datasets.
- Developed an asynchronous API harness via the DeepSeek API and a unified evaluation engine to test zero-shot baselines, including Chain-of-Thought, Self-Consistency, and Self-Refine.
- Conducted quantitative analysis on LLM hallucination and "Degeneration-of-Thought," establishing Self-Consistency as the optimal token-efficient inference strategy for deterministic tasks.

Financial 10-K Orchestration Agent | *Python, LLM API, ChromaDB, Streamlit, pdfplumber*

May 2026 - Jun 2026

- Architected a layout-aware RAG pipeline for SEC 10-K filings using an embedded ChromaDB vector store, engineering a metadata fan-out retrieval strategy that achieved 100% data inclusion for queries and eliminated context starvation.
- Developed a two-pass agentic orchestration layer utilizing DeepSeek-V4-Flash at temperature 0.0, implementing a deterministic intent router for conversational query rewriting, coreference resolution, and strict target filtering.
- Built a reactive Streamlit interface featuring asynchronous real-time token streaming, transparent source-citation auditing, and a proactive "Next Best Action" suggestion engine to guide user data discovery.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, Java, C

AI & Machine Learning: LLM Orchestration, RAG Frameworks, Prompt Engineering, FAISS, Scikit-learn, Hugging Face

Cloud & Data Engineering: Snowflake, Azure Data Factory (ADF), AWS, ETL Pipelines

Databases: MySQL, MariaDB, SQLite, MongoDB

Tools & Platforms: Git, GitHub, Docker, REST APIs, OpenRouter, CI/CD, Jira, Power BI